GastroOps: System Evaluation & Quality Audit Report

DeepEval Metric Verification • End-to-End Latency Benchmarks • LLM-as-Judge Audit

1. Executive Quality Assurance Summary

GastroOps employs a rigorous, automated multi-layered quality evaluation strategy. To guarantee production reliability and eliminate generative hallucinations, recommendations and forensic diagnoses are audited across four deterministic DeepEval dimensions, qualitative LLM-as-Judge rubric audits, and an automated 11-point pytest integration test suite.

Quality Dimension	Evaluation Target	Weight	Pass Threshold
Faithfulness	Factual consistency & citation grounding against real Yelp reviews	30%	Score ≥ 0.70
Answer Relevancy	Direct alignment with the manager's queried operational failure scenario	30%	Score ≥ 0.70
Aspect Coverage	Completeness across all diagnosed primary friction failure points	20%	Score ≥ 0.60
Operational Feasibility	Actionable SOP codes (SOP-FQ-101), time limits, and measurable KPIs	20%	Score ≥ 0.75

2. DeepEval Empirical Benchmark Results

The following empirical results were recorded across diverse operational stress scenarios on the live GastroOps production server:

Scenario Query	Faithfulness	Relevancy	Coverage	Feasibility	Composite	Status
Kitchen ticket delays and undercooked chicken entrees	0.88	0.89	0.67	0.98	**86.1%**	PASSED
Slow service and cold food	0.88	0.94	0.67	0.98	**87.6%**	PASSED
Rude server and inattentive attitude	0.84	0.91	0.67	0.98	**85.5%**	PASSED
What are the good reviews (Commendation)	0.84	0.86	0.67	0.98	**84.0%**	PASSED
Overpriced food and small portion sizes	0.77	0.74	0.67	0.98	**78.3%**	PASSED

3. LLM-as-Judge Qualitative Rubric Audit

Agent 6 (Executive Auditor & LLM-as-Judge Agent) executes a qualitative rubric evaluation prior to publishing action plans. Each plan is independently graded on a 1.0–5.0 point scale:

Rubric Dimension	Score	Evaluation Finding
Evidence Grounding	4.2 / 5.0	High correlation with guest verbatims; zero fabricated operational grievances.
Root-Cause Precision	4.7 / 5.0	Accurately isolated pass-window temperature decay and sauté pacing breakdowns.

Actionability	4.2 / 5.0	Enforces 2 distinct preventative SOPs and 2 immediate 24h corrective controls.
Severity Calibration	4.4 / 5.0	Priority Tier correctly mapped to clinical/safety thresholds (P0-P4).

4. End-to-End Latency Profile & Performance Benchmarks

Pipeline Stage	Avg Latency	Optimization Method
Input Validation & Guardrail	2.4 ms	Regex & compiled boundary security filters
Query Understanding & Routing	0.2 ms	Local in-memory semantic router dictionary
Hybrid Dense + Sparse Retrieval	180 - 280 ms	Qdrant HNSW indexing + local BM25 in-memory tokens
BGE Cross-Encoder Reranking	45 - 80 ms	Pre-filtered 6-candidate pool inference
Multi-Agent Collaborative Core	12 - 25 ms	Deterministic rule & taxonomy execution
Executive Review Synthesis (LLM)	450 - 750 ms	NVIDIA NIM / Fallback dynamic local NLP
DeepEval Automated Audit	< 8 ms	Pre-tokenized in-memory rubric verification
Multilingual Translation	280 ms (1st call) / 0 ms (cached)	Dual-tier React & backend in-memory cache
Total End-to-End Latency	**0.92 - 1.15 s**	**Sub-second interactive execution**

5. Automated Regression Test Suite (pytest)

The project maintains **11 integration test cases** in backend/tests/test_api_endpoints.py with a **100% pass rate (11/11 PASSED)** covering health, config, business metadata, hybrid search, CSAT regression bounds, time-series slope trends, feedback storage, multi-agent pipeline, language endpoints, multilingual translation preservation, and positive review query commendation routing.